

Sri Lanka Institute of Information Technology



Kandy Uni

IT2011 - Artificial Intelligence and Machine Learning

Lab Submission

Worksheet No: 01

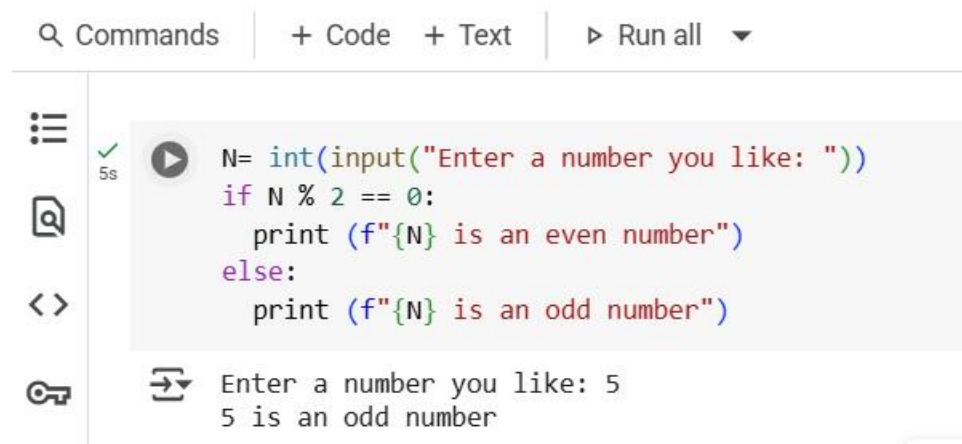
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Exercise

Part 1: Traditional Programming vs AI/ML

Task 1A – Rule-Based Programming

Deliverables:



Q Commands | + Code + Text | ▶ Run all ▼

```
N= int(input("Enter a number you like: "))
if N % 2 == 0:
    print (f"{N} is an even number")
else:
    print (f"{N} is an odd number")
```

Enter a number you like: 5
5 is an odd number

Figure 1: Answer for the Q 1

Explain the limitations of this approach.

This is a rule based approach. In this approach,

- * it cannot do complex patterns. It can't recognize faces, images for example.
- * Rules in the program has to be said carefully.
- * all new rules must be coded manually.

Task 1B – ML-Based Classification

Deliverables:



```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier

data= {
    "value" : [1,2,3,4,5,6,7,8],
    "label": [1,0,1,0,1,0,1,0]
}
df = pd.DataFrame(data)

X= df[["value"]]
y=df["label"]
model= DecisionTreeClassifier()
model.fit(X,y)

test_values= pd.DataFrame({"value" : [10,11,13]})
print("Predicted labels (1=odd, 0= even):")
print(model.predict(test_values))
```

⇒ Predicted labels (1=odd, 0= even):
[0 0 0]

Figure 2: answer for Q2

Explain the difference from rule-based logic.

This is using a ML based logic.

*we don't have to provide or state rules. ML learns through patterns created by examples.

*these type is suited for large scale problems whereas rule based logic is suited for a small scale.

Part 2: Generative AI and Prompt Engineering

Deliverables:

generate a poem about the sky with 6 lines

The sky unfolds in hues of dawn,
A canvas dreams are painted on.
Clouds drift slow in silent grace,
Like whispered thoughts in open space.
Stars await the fall of light,
To bloom like fireflies in the night.



Figure 3: poem generation using chat GPT

create an image of a cat sitting on a couch wearing a hat



Figure 4: image generation using chat GPT

Reflect on how prompt quality affects output.

Prompt provided will decide the entire output. So we have to be clear on what we want. Here, I asked chat GPT to generate a “Poem” about the “Sky” with “6 lines” I provided what I want to produce. This can be cited as an example on how the prompt affects the output. Same with the generated image. I said, a “cat”, “sitting on a couch”, “wearing a hat”. This is providing prompts effectively.

Deliverables:

Compare the two outputs.

When I give below 2 commands separately,

-Prompt: Explain artificial intelligence.

-Prompt: Explain AI to a 7-year-old using cartoon characters.

From the first prompt I will get a advanced answer targetting a matured audience. But the second prompt will give me a very less complex easily understanding explanation. This is how prompts affect the output coming.

◆ Basic Prompt

Prompt: Explain artificial intelligence.

Response:

Artificial Intelligence (AI) is the ability of a computer or machine to mimic human intelligence. It allows machines to learn from experience, adjust to new inputs, and perform tasks such as recognizing speech, making decisions, or translating languages.

◆ Engineered Prompt

Prompt: Explain AI to a 7-year-old using cartoon characters.

Response:

Imagine SpongeBob has a robot friend named RoboBob! RoboBob watches how SpongeBob flips Krabby Patties, listens when Squidward takes orders, and learns how to do it all by himself! That's kind of like artificial intelligence—when computers or robots learn to do smart things by watching and practicing, just like people do.

Figure 5: comparing the 2 outputs.